Assignment-7

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Logistic Regression for Binary Classification

Objective

1. Understand logistic regression for classification problems.
2. Build a logistic regression model in R.
3. Predict probabilities and classes.
4. Evaluate the model using classification metrics.

Dataset

For logistic regression we need a **binary target variable**.

We will convert the **Advertising dataset** into a classification problem.

Rule:

```
If Sales > median(Sales) → High Sales Else → Low Sales
```

```r
library(caret)
```

```
Loading required package: ggplot2

Loading required package: lattice
```

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
#Load DataSet
setwd("D:/Sem-2/R-/7,8,9,10/7")
data<-read.csv("Advertising.csv")

head(data)
```

```
  X    TV radio newspaper sales
1 1 230.1  37.8      69.2  22.1
2 2  44.5  39.3      45.1  10.4
3 3  17.2  45.9      69.3   9.3
4 4 151.5  41.3      58.5  18.5
5 5 180.8  10.8      58.4  12.9
6 6   8.7  48.9      75.0   7.2
```

Creating binary target variable

```r
median_sales <- median(data$sales)

data$Sales_Class <- ifelse(data$sales > median_sales, 1, 0)

head(data)
```

```
  X    TV radio newspaper sales Sales_Class
1 1 230.1  37.8      69.2  22.1           1
2 2  44.5  39.3      45.1  10.4           0
3 3  17.2  45.9      69.3   9.3           0
4 4 151.5  41.3      58.5  18.5           1
5 5 180.8  10.8      58.4  12.9           0
6 6   8.7  48.9      75.0   7.2           0
```

Convert to Factor

```
data$Sales_Class <- as.factor(data$Sales_Class)
```

Train Test Split

```
set.seed(123)

train_index <- createDataPartition(data$Sales_Class,
                                   p = 0.7,
                                   list = FALSE)

train_data <- data[train_index, ]
test_data <- data[-train_index, ]

head(train_data)
```

```
    X    TV radio newspaper sales Sales_Class
1   1 230.1  37.8      69.2  22.1           1
6   6   8.7  48.9      75.0   7.2           0
7   7  57.5  32.8      23.5  11.8           0
9   9   8.6   2.1       1.0   4.8           0
10 10 199.8   2.6      21.2  10.6           0
11 11  66.1   5.8      24.2   8.6           0
```

```
head(test_data)
```

```
    X    TV radio newspaper sales Sales_Class
2   2  44.5  39.3      45.1  10.4           0
3   3  17.2  45.9      69.3   9.3           0
4   4 151.5  41.3      58.5  18.5           1
5   5 180.8  10.8      58.4  12.9           0
8   8 120.2  19.6      11.6  13.2           1
14 14  97.5   7.6       7.2   9.7           0
```

Build Logistic Regression Model

Model equation

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where

$$z = \beta_0 + \beta_1 TV + \beta_2 Radio + \beta_3 Newspaper$$

```r
model_logistic <- glm(Sales_Class ~ TV + radio + newspaper,
                      data = train_data,
                      family = binomial)

summary(model_logistic)
```

```
Call:
glm(formula = Sales_Class ~ TV + radio + newspaper, family = binomial,
    data = train_data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -16.821312   3.861133  -4.357 1.32e-05 ***
TV            0.062432   0.014039   4.447 8.71e-06 ***
radio         0.327794   0.077539   4.227 2.36e-05 ***
newspaper    -0.001958   0.019084  -0.103    0.918
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 195.404  on 140  degrees of freedom
Residual deviance:  44.127  on 137  degrees of freedom
AIC: 52.127

Number of Fisher Scoring iterations: 8
```

Predict Probabilities

Logistic regression outputs **probabilities**.

```
probabilities <- predict(model_logistic,
                         test_data,
                         type = "response")

head(probabilities)
```

```
           2            3            4            5            8           14
0.2228516879 0.3020717130 0.9976719527 0.1083335120 0.0514169548 0.0002593413
```

Convert Probabilities to Classes

- Rule:

  `Probability > 0.5 → Class 1 Else → Class 0`

```
predicted_class <- ifelse(probabilities > 0.5, 1, 0)

predicted_class <- as.factor(predicted_class)
```

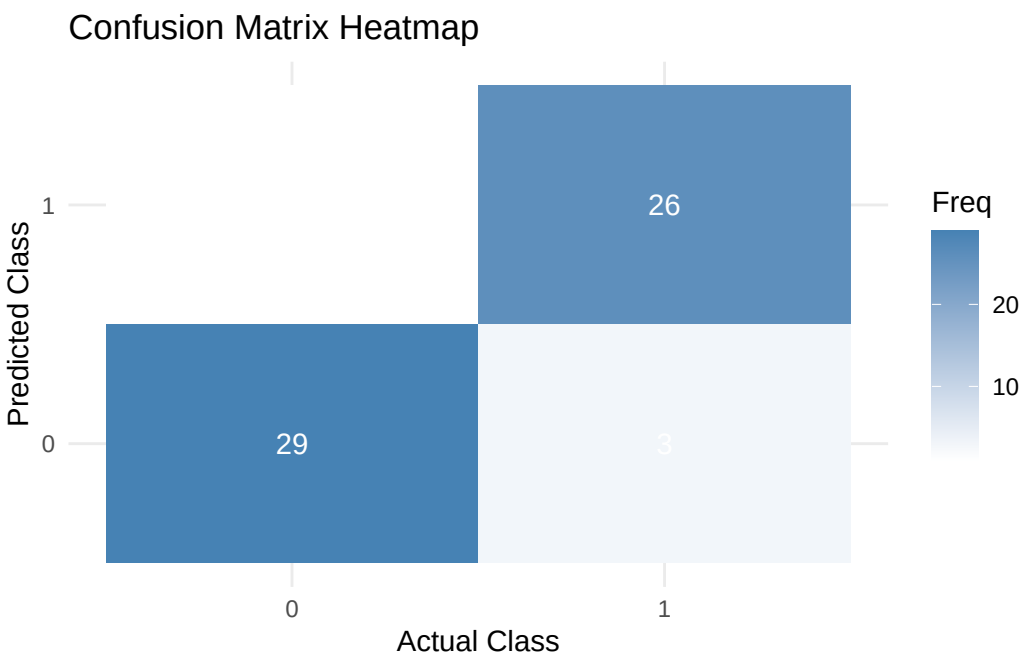
Confusion Matrix

```
conf_matrix <- confusionMatrix(predicted_class,
                               test_data$Sales_Class)

conf_matrix
```

```
Confusion Matrix and Statistics

          Reference
Prediction  0  1
         0 29  3
         1  1 26

               Accuracy : 0.9322
                 95% CI : (0.8354, 0.9812)
```

```
    No Information Rate : 0.5085
    P-Value [Acc > NIR] : 2.003e-12

                  Kappa : 0.8642

 Mcnemar's Test P-Value : 0.6171

            Sensitivity : 0.9667
            Specificity : 0.8966
         Pos Pred Value : 0.9062
         Neg Pred Value : 0.9630
             Prevalence : 0.5085
         Detection Rate : 0.4915
   Detection Prevalence : 0.5424
      Balanced Accuracy : 0.9316

       'Positive' Class : 0
```

```r
library(ggplot2)

# Convert matrix to a data frame
df <- as.data.frame(conf_matrix$table)

# Plot as a heatmap
ggplot(df, aes(x = Reference, y = Prediction, fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = Freq), color = "white") +
  scale_fill_gradient(low = "white", high = "steelblue") +
  labs(title = "Confusion Matrix Heatmap", x = "Actual Class", y = "Predicted Class") +
  theme_minimal()
```



Accuracy

Accuracy measures overall correctness.

$$Accuracy = \frac{TP + TN}{Total}$$

```
conf_matrix$overall['Accuracy']
```

```
 Accuracy
0.9322034
```

Precisioon

Precision measures how many predicted positives are correct.

$$Precision = \frac{TP}{TP + FP}$$

```r
conf_matrix$byClass['Precision']
```

```
Precision
  0.90625
```

Recall

Recall measures how many actual positives are detected.

$$Recall = \frac{TP}{TP + FN}$$

```r
conf_matrix$byClass['Recall']
```

```
   Recall
0.9666667
```

F1-Score

F1 combines precision and recall.

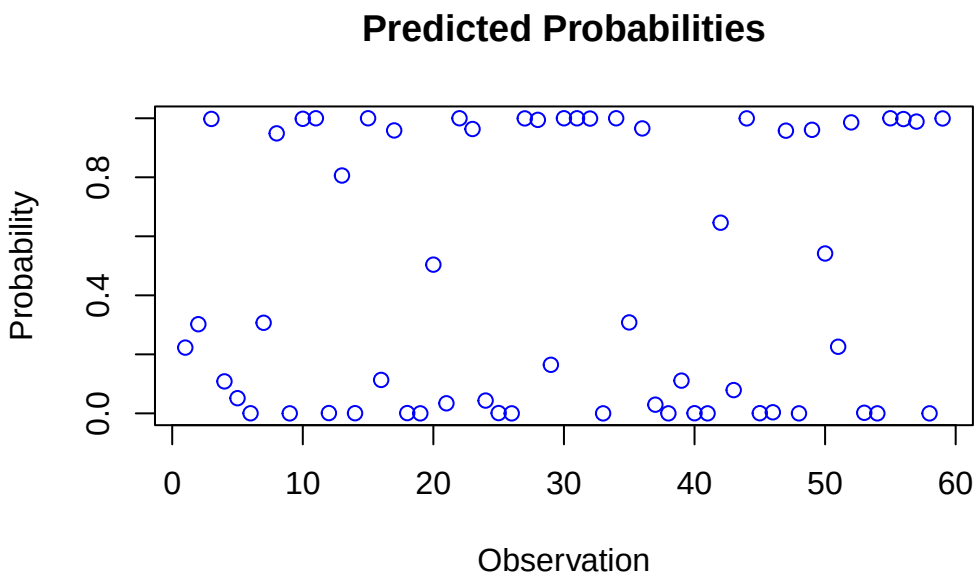
$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

```r
precision <- conf_matrix$byClass['Precision']
recall <- conf_matrix$byClass['Recall']

F1 <- 2 * (precision * recall) / (precision + recall)

F1
```

```
Precision
0.9354839
```

Visualizing Class Predictions

```r
plot(probabilities,
     col="blue",
     main="Predicted Probabilities",
     ylab="Probability",
     xlab="Observation")
```



Simple and Multiple Linear Regression using Advertising Dataset

Loading the libraries

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.4     v tibble    3.3.1
v purrr     1.2.1     v tidyr     1.3.2
v readr     2.1.6
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
```

```
x purrr::lift()   masks caret::lift()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(GGally)
library(caret)
```

```r
#understanding data set
str(data)
```

```
'data.frame':   200 obs. of  6 variables:
 $ X          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ TV         : num  230.1 44.5 17.2 151.5 180.8 ...
 $ radio      : num  37.8 39.3 45.9 41.3 10.8 48.9 32.8 19.6 2.1 2.6 ...
 $ newspaper  : num  69.2 45.1 69.3 58.5 58.4 75 23.5 11.6 1 21.2 ...
 $ sales      : num  22.1 10.4 9.3 18.5 12.9 7.2 11.8 13.2 4.8 10.6 ...
 $ Sales_Class: Factor w/ 2 levels "0","1": 2 1 1 2 1 1 1 2 1 1 ...
```

```r
summary(data)
```

```
       X                TV             radio          newspaper
 Min.   :  1.00   Min.   :  0.70   Min.   : 0.000   Min.   :  0.30
 1st Qu.: 50.75   1st Qu.: 74.38   1st Qu.: 9.975   1st Qu.: 12.75
 Median :100.50   Median :149.75   Median :22.900   Median : 25.75
 Mean   :100.50   Mean   :147.04   Mean   :23.264   Mean   : 30.55
 3rd Qu.:150.25   3rd Qu.:218.82   3rd Qu.:36.525   3rd Qu.: 45.10
 Max.   :200.00   Max.   :296.40   Max.   :49.600   Max.   :114.00
     sales       Sales_Class
 Min.   : 1.60   0:102
 1st Qu.:10.38   1: 98
 Median :12.90
 Mean   :14.02
 3rd Qu.:17.40
 Max.   :27.00
```

Scatter Plot Matrix

```r
ggpairs(data)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
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```
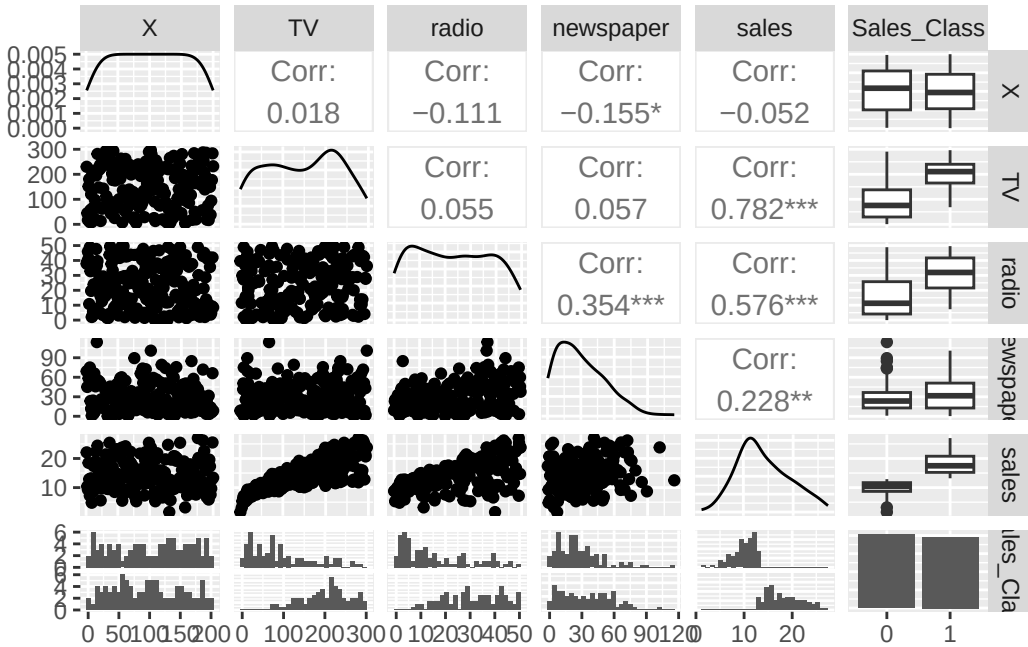



Simple Linear Regression

We Predict Sales using only TV Advertising.

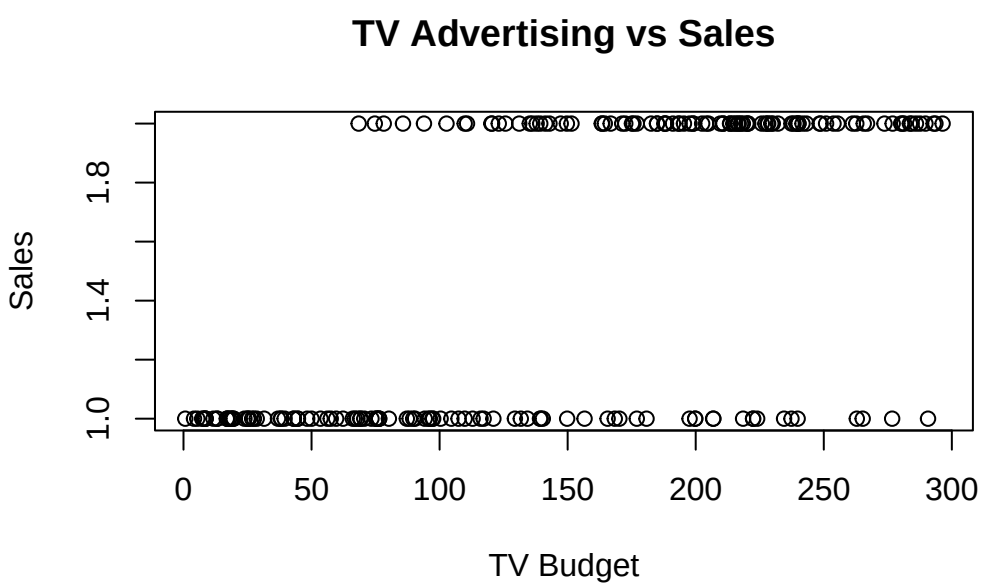
```r
model_simple <- lm(sales ~ TV, data = data)

summary(model_simple)
```

```
Call:
lm(formula = sales ~ TV, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.3860 -1.9545 -0.1913  2.0671  7.2124 
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.032594   0.457843   15.36   <2e-16 ***
TV          0.047537   0.002691   17.67   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.259 on 198 degrees of freedom
Multiple R-squared:  0.6119,    Adjusted R-squared:  0.6099
F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

Plotting The regression line

```r
plot(data$TV, data$Sales,
     main = "TV Advertising vs Sales",
     xlab = "TV Budget",
     ylab = "Sales")

abline(model_simple, col="red", lwd=2)
```



Multiple Linear Regression

Now using all advertising channels

```r
model_multiple <- lm(sales ~ TV + radio + newspaper, data = data)

summary(model_multiple)
```

```
Call:
lm(formula = sales ~ TV + radio + newspaper, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.8277 -0.8908  0.2418  1.1893  2.8292 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.938889   0.311908   9.422   <2e-16 ***
TV           0.045765   0.001395  32.809   <2e-16 ***
radio        0.188530   0.008611  21.893   <2e-16 ***
newspaper   -0.001037   0.005871  -0.177     0.86    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.686 on 196 degrees of freedom
Multiple R-squared:  0.8972,	Adjusted R-squared:  0.8956 
F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

Prediction

- Predict sales using the model

```r
new_data <- data.frame(TV=150, radio=25, newspaper=20)

predict(model_multiple, new_data)
```

```
       1 
14.49609 
```

Model Evaluation

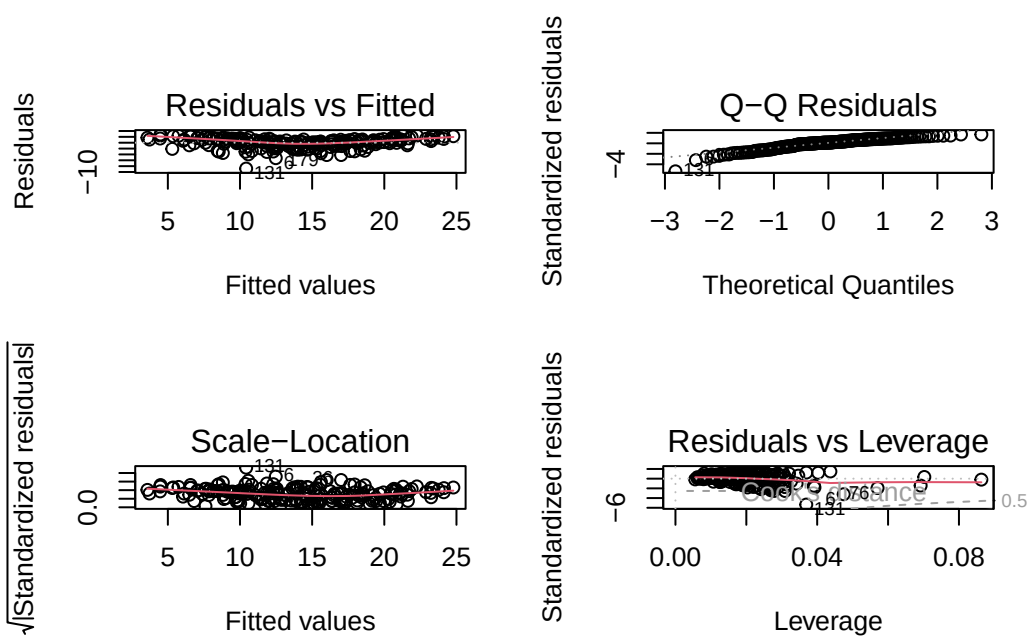
- Shows how much variation in Sales is explained by predictors.

```
summary(model_multiple)$r.squared
```

```
[1] 0.8972106
```

```
pred <- predict(model_multiple, data)

RMSE(pred, data$sales)
```

```
[1] 1.66857
```

Residual Analysis

```
par(mfrow=c(2,2))
plot(model_multiple)
```